143-1

[redacted]

INVOICE CHECK LIST

MULTRA Subproject

At Date of Original Authorization	Period Covered	Time Extended To	Allotment Number	Amount of Obligation
11 Dec 1962			3125-1390-3902	20,629.44

Additional Authorizations	Period Covered	Time Extended To	Allotment Number	Amount of Obligation
25 Jan 63			3125-1390-3902	20,800.00

Invoice Number	Date	Amount	Balance
Auth #1	11 Dec 1962	20,629.44	20,629.44
		19836.00	
Rec #1		793.44	-0-
Auth #2	25 Jan 63	20,800.00	20,800.00
		800.00	
Inv #2	25 Jan 63	20,000.00	-0-

REMARKS:

[redacted]

143-7

(When Filled In)

ACCOUNTING BY INDIVIDUAL FOR ADVANCE

NOTE: Follow Instructions on Reverse

SUBMITTED BY: MKULTRA Sub-project # 143

PERIOD OF ACCOUNTING: FROM 1 January 1963 TO 22 October 1965

DATE - VOUCHER NO. 2-12

1. CASH ON HAND BEGINNING OF PERIOD: $ 39,836.00
2. OUTSTANDING ADVANCES BEGINNING OF PERIOD:
3. RECEIPTS THIS PERIOD:

RECEIPT NUMBER	DATE	DESCRIPTION

5. EXPENSES THIS PERIOD:

VOUCHER NUMBER	DATE	DESCRIPTION	AMOUNT
		Expenses - See attached certification	$ 39,836.00
		TOTAL EXPENSES	$ 39,836.00

6. REFUNDED HEREWITH: CASH ( ) CHECK ( ) MONEY ORDER ( )
7. OUTSTANDING ADVANCES END OF PERIOD (Attach listing)
8. CASH ON HAND END OF PERIOD OR BALANCE DUE ADVANCEE
9. TOTAL ACCOUNTED FOR: $ 39,836.00

4. TOTAL TO ACCOUNT FOR: $ 39,836.00

I CERTIFY FUNDS ARE AVAILABLE

OBLIG. REF. NO.	CHARGE FAN ACCOUNT NO.	
1016	3125-1390-3902	$19,836.00
400	4125-1390-3902	$20,000.00

DATE:

SIGNATURE OF AUTHORIZING OFFICER:

APPROVED

DATE:

SIGNATURE OF APPROVING OFFICER: SIDNEY GOTTLIEB, C/TSD

CERTIFIED FOR PAYMENT OR CREDIT

DATE:

SIGNATURE OF CERTIFYING OFFICER:

I certify that the expenditures listed hereon and on any attachments were incurred for official purposes of a confidential nature, that payment or credit therefor has not been received, and that this accounting is true and correct.

SIGNATURE OF PAYEE: A [redacted] C/TSD/BB

SPACE BELOW FOR EXCLUSIVE USE OF OFFICE OF FINANCE

DESCRIPTION - ADVANCE ACCOUNTS 13-27	DESCRIPTION - ALL OTHER ACCOUNTS 13-33	28-33 Y/A NO. / SHIP. DOC. NO.	34-39 STATION CODE / REC. APT. NO. / PROJECT NO.	40-41 EXPEND CODE	42-47 OBLIG. REF. NO. / ADVANCE ACCT. NO. / EMP. NO.	48-49 PAY PER. LIQ. CODE	50 CA OR COST YR.	51-54 GENERAL LEDGER ACCT. NO.	55-66 COST - FAN ACCOUNT SYMBOL	61-66 CR. NO. / X REF. NO.	67-70 OBJECT CLASS / 68-70 DUE DATE	71-80 AMOUNT DEBIT	71-80 AMOUNT CREDIT
INV. 1 MKA [illegible]		25	77-185		143			7100					19,836.00
INV. 2 MKA [illegible]		25	77-185		143			7100					20,000.00
MKA INV 1 2								7990				39,836.00	

DATE: 17 Nov 196[illegible]

PREPARED BY: [redacted] A

DATE:

REVIEWED BY: [redacted]

TOTALS: 39,836.00 | 39,836.00

FORM 4-65 282 OBSOLETE PREVIOUS EDITIONS

GROUP 1 EXCLUDED FROM AUTOMATIC DOWNGRADING AND DECLASSIFICATION

(1-49)

143-3

[redacted]

CERTIFICATION

This is to certify that I have received an accounting from MKULTRA Sub-project # 143 for the period 1 January 1963 - 22 October 1965. The accounting reflects expenses in the amount of $39,823.29 against the remaining grant balance of $39,836.00. The residual balance of $12.71 should be written off as being impracticle for collection or detailed accounting.

The accounting statement and payment certification will be retained by TSD and will be made available for review in TSD if necessary.

I certify that satisfactory services represented by the accounting have been received and that to the best of my knowledge and belief the funds expended were for the purposes authorized by the project approval.

[redacted]
Chief, TSD/EB

[redacted]

[redacted]

143-9

[redacted]
Statement of Expenditures
[redacted]

	Expenditures 1/1/63-10/22/65
Salaries	
[redacted]	$ 9,089.17
	650.00
	7,710.00
	400.00
	1,995.00
	250.00
	1,458.00
	2,150.00
	2,180.00
	200.00
	75.00
Total Salaries	26,157.17
Consumable Supplies	8.34
F I C A	5.55
Telephone & Telegram	575.48
Tubes & pipettes	1,066.35
Misc. Supplies	115.93
Electrical Supplies	176.07
Chemicals	15.33
Freight bills	113.16
Filter apparatus	599.71
Cultures	357.99
Flasks, stoppers, syringes, etc.	89.65
Repair on Vacuum pump	59.50
Test tube mixer	120.00
Service on a Sartorius Balance	3,303.11
Total Consumable Supplies	
Travel	
[redacted] 7/31/63-8/6/63,	457.04
[redacted]	27.90
[redacted] 4/3/64-4/4/64,	21.82
[redacted] 10/19/64-10/24/64,	174.25

Continued on next page.

4-63

143-5

[redacted] (When Filled In) No. 400

Cost Account 4125-1350-3902

Object Class

Date	Remarks and References	Obligations Incurred	Obligations Liquidated	Unliquidated Balance
1/30/64	MKULTRA SUB 143	20,000.00		20,000.00

[redacted] (When Filled In)

143-6

(When Filled In)

REQUEST FOR ADVANCE OF FUNDS

NAME OF ADVANCEE	DATE	AMOUNT	VOUCHER NO. (Finance use only)
[redacted]	30 January 1964	$800.00 $20,000.00	
PAYABLE TO	ROOM / BUILDING	TELEPHONE EXT.	
See attached payment inst. B-10	[redacted]	[redacted]	

PURPOSE

Advance to [redacted] for funding MKULTRA Subproject 143, invoice # 2 which activity was approved by C/TSD on 25 January 1964. Accounting to be in accordance with MKULTRA Fiscal Annex attachment B.

Please forward checks (2) to C/TSD/DB thru TSD/BF no later than 10 February 1964.

I agree that I will fully account for this advance by submission of vouchers and refund of any unexpended balance to the reporting point stated and by the due date checked below. In the event of my failure to so account and refund any unexpended balance, I authorize deduction from my salary to effect settlement.

STATUS OF OUTSTANDING ADVANCES

DATE	AMOUNT	UNACCOUNTED BALANCE

REPORTING POINT

FINANCE DIVISION - HEADQUARTERS

DUE DATE ✓

075

ON ARRIVAL AT DESTINATION ON OR ABOUT

MONTHLY - ON THE LAST WORKDAY OF EACH MONTH

REQUESTING OFFICER

DATE: 1/30/64

SIGNATURE: [redacted]

I CERTIFY FUNDS ARE AVAILABLE

OBLIGATION REFERENCE NO.: 400

CHARGE ALLOTMENT NO.: 4125-1390-3902

DATE

SIGNATURE

APPROVED

DATE

SIGNATURE OF APPROVING OFFICER

SIDNEY GOTTLIEB, DC/TSD

CERTIFIED FOR PAYMENT OR CREDIT

DATE

AUTHORIZED CERTIFYING OFFICER

RECEIPT FOR FUNDS ADVANCED

I acknowledge receipt of funds in the amount stated hereunder to be used for the purpose stated and accounted for as shown above.

DATE

AMOUNT

SIGNATURE OF ADVANCEE

SPACE BELOW FOR EXCLUSIVE USE OF FINANCE DIVISION

PREPARED BY

REVIEWED BY

VOUCHER NO. 7-12

DESCRIPTION-ALL OTHER ACCOUNTS 13-33	28-33 T/A NO.	34-39 STATION CODE	40-42 EXPEND CODE	43 FUNDS	45-46 PAY PERIOD CODE	47-52 OBLIG. REF. NO.	53 CA YR	54-57 GENERAL LEDGER ACCT. NO.	58-67 ALLOT. OR COST ACCT. NO.	62-67 CK. NO. / REF. NO.	68-70 DUE DATE	71-80 AMOUNT DEBIT	AMOUNT CREDIT
DESCRIPTION-ADVANCE ACCOUNTS 13-27	P.O. NO. / 32-33 DIV.	PROP. NO. / PROJECT NO.	IFY			ADVANCE ACCT. NO. / EMP. NO.					OBJECT CLASS		
[redacted]	[redacted]	[redacted]	[redacted]	[redacted]	[redacted]	[redacted]	[redacted]	[redacted]	[redacted]	[redacted]	[redacted]	800.00	
										TOTALS			

FORM 5-60 281 USE PREVIOUS EDITIONS.

* U.S. GOVERNMENT PRINTING OFFICE : 1962 O—649743

(38-49)

143-7

143



B

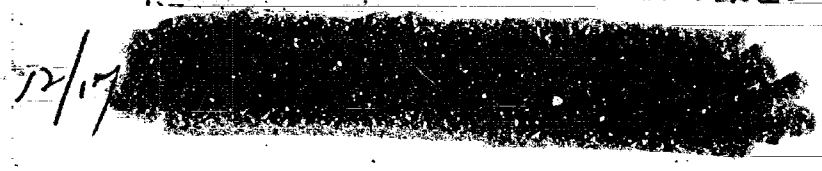
Tiller

E

E

CHEC[illegible] AMOUNT OF 180 00

12/17



A

143

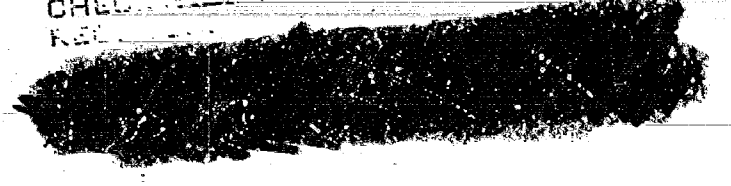


B

E

E

CHEC[illegible] AMOUNT OF 4600 00



A

12/17

143-8

DATE: 8 January 1964

MEMORANDUM FOR : C/TSD/FASS

SUBJECT : MKULTRA, Subproject No. 143

Under the authority granted in the memorandum dated 13 April 1953, from the DCI to the DD/A, and the extension of this authority in subsequent memoranda, Subproject 143 has been approved, and $20,800.00 of the over-all Project MKULTRA funds have been obligated to cover the subproject's expenses and should be charged to cost center 4125-1390-3902

[redacted]

A

PROGRAM MANAGER

Concur:

13 JAN 1964

Asst. Chief, TSD for D&E

Concur:

Category IV a

APPROVED FOR OBLIGATION OF FUNDS:

Chief, DD/P/TSD

Date:

Distribution:
Orig & 1 - Addressee
2 - TSD/
BB

GROUP I
Excluded from automatic downgrading and declassification

[redacted] (When Filled In)

No. 1016

Cost Account 3125-1390-3902

Object Class

Date	Remarks and References	Obligations Incurred	Obligations Liquidated	Unliquidated Balance
18 DEC 1962	MKULTRA Sub Proj. 143 - Auth #1	20,629.44		20,629.44
26 DEC 1962	Invoice #1		20,629.44	

[redacted] (When Filled In)

143-9

Date: 11 December 1962

MEMORANDUM FOR: C/TSD/FASS

SUBJECT : MKULTRA, Subproject No. 143
Authorization # 1

Under the authority granted in the memorandum dated 13 April 1953, from the DCI to the DD/A, and the extension of this authority in subsequent memoranda, Subproject 143 has been approved and $20,629.44 of the over-all Project MKULTRA funds have been obligated to cover the subproject's expenses and should be charged to cost center 3125-1390-3902.

[redacted]

Concur:

[redacted]
Asst. Chief, TSD for

A

Concur:

[redacted]

APPROVED FOR OBLIGATION OF FUNDS:

[redacted]
DD/

Date: 18 Dec 1962

I CERTIFY THAT FUNDS ARE AVAILABLE
OBLIGATION [illegible] 1016
CHARGE [illegible] 90-3902

[redacted]

Distribution:
Orig & 1 - Addressee
2 - TSD/

GROUP 1
Excluded from automatic downgrading and declassification

REQUEST FOR ADVANCE OF FUNDS

(When Filled In)

NAME OF ADVANCEE	DATE	AMOUNT	VOUCHER NO. (Finance use only)
[redacted]	19 December '62	$19,836.00 + 793.44 (Fee)	
PAYABLE TO: See attached Payment Instr.	ROOM: 119 / BUILDING: [redacted]	TELEPHONE EXT.: [redacted]	

PURPOSE

Advance to NEULTRA Subproject 143 Invoice # 1 for activity approved by C/TSD on 27 December 1962. Accounting for this advance will be in accordance with attachment A of NEULTRA Fiscal Annex.

Please forward the check to Chief,TSD/Biological Branch through TSD/Budget Officer by 3 January 1963.

143-9

STATUS OF OUTSTANDING ADVANCES

DATE	AMOUNT	UNACCOUNTED BALANCE

I agree that I will fully account for this advance by submission of vouchers and refund of any unexpended balance to the reporting point stated and by the due date checked below. In the event of my failure to so account and refund any unexpended balance, I authorize deduction from my salary to effect settlement.

REPORTING POINT	DUE DATE	✓
FINANCE DIVISION - HEADQUARTERS		
	ON ARRIVAL AT DESTINATION	
	ON OR ABOUT	
	MONTHLY - ON THE LAST WORKDAY OF EACH MONTH	

REQUESTING OFFICER

DATE | SIGNATURE

APPROVED

DATE | SIGNATURE OF APPROVING OFFICER: SIDNEY GOTTLIEB AC/TSD

RECEIPT FOR FUNDS ADVANCED

I acknowledge receipt of funds in the amount stated hereunder to be used for the purpose stated and accounted for as shown above.

DATE | AMOUNT

I CERTIFY FUNDS ARE AVAILABLE

OBLIGATION REFERENCE NO.: 1016 | CHARGE ALLOTMENT NO.: 3125-1390-3902

DATE | SIGNATURE

CERTIFIED FOR PAYMENT OR CREDIT

DATE | AUTHORIZED CERTIFYING OFFICER

SIGNATURE OF ADVANCEE

SPACE BELOW FOR EXCLUSIVE USE OF FINANCE DIVISION

PREPARED BY | REVIEWED BY: [redacted] | VOUCHER NO. 7-12

DESCRIPTION-ADVANCE ACCOUNTS 13-27	DESCRIPTION-ALL OTHER ACCOUNTS 13-33	28-33 T/A NO. P.O. NO. 32-33 DIV.	34-39 STATION CODE PROP. NO. PROJECT NO.	40-42 EXPEND CODE FY	43 FUNDS	45-46 PAY PER. LIQ. CODE	47-52 OBLIG. REF. NO. ADVANCE ACCT. NO. EMP. NO.	53 CAL YR	54-57 GENERAL LEDGER ACCT. NO.	58-67 OR C/T NO.	68-70 DUE DATE	71-80 AMOUNT DEBIT	71-80 AMOUNT CREDIT
											TOTALS		

CHECK #[redacted] 1/4/63

FORM 5-60 281 USE PREVIOUS EDITIONS. (38-49)

(When Filled In)

REQUEST FOR ADVANCE OF FUNDS

NAME OF ADVANCEE	DATE	AMOUNT	VOUCHER NO. (Finance use only)
[redacted]	19 December '62	$19,836.00 + $ 793.44 (Fee)	
PAYABLE TO	**ROOM** / **BUILDING**	**TELEPHONE EXT.**	
See attached Payment Inst.	119 / [redacted]	[redacted]	

PURPOSE

Advance to MKULTRA Subproject 143 Invoice # 1 for activity approved by C/TSD on 17 December 1962. Accounting for this advance will be in accordance with attachment A of MKULTRA Fiscal Annex.

Please forward the check to Chief,TSD/Biological Branch through TSD/Budget Officer by 3 January 1963.

143

STATUS OF OUTSTANDING ADVANCES

DATE	AMOUNT	UNACCOUNTED BALANCE

I agree that I will fully account for this advance by submission of vouchers and refund of any unexpended balance to the reporting point stated and by the due date checked below. In the event of my failure to so account and refund any unexpended balance, I authorize deduction from my salary to effect settlement.

REPORTING POINT

FINANCE DIVISION - HEADQUARTERS

DUE DATE ✓

- 065 ✓
- ON ARRIVAL AT DESTINATION
- ON OR ABOUT
- MONTHLY - ON THE LAST WORKDAY OF EACH MONTH

REQUESTING OFFICER

DATE | SIGNATURE

APPROVED

DATE | SIGNATURE OF APPROVING OFFICER: SIDNEY GOTTLIEB AC/TSD

RECEIPT FOR FUNDS ADVANCED

I acknowledge receipt of funds in the amount stated hereunder to be used for the purpose stated and accounted for as shown above.

DATE | AMOUNT | SIGNATURE OF ADVANCEE

I CERTIFY FUNDS ARE AVAILABLE

OBLIGATION REFERENCE NO.: 1016 | CHARGE ALLOTMENT NO.: 3125-1390-3902

DATE | SIGNATURE

CERTIFIED FOR PAYMENT OR CREDIT

DATE | AUTHORIZED CERTIFYING OFFICER

SPACE BELOW FOR EXCLUSIVE USE OF FINANCE DIVISION

REVIEWED BY [redacted] 12/6[illegible]

VOUCHER NO. 7-12 [illegible]

TYPE	DESCRIPTION - ADVANCE ACCOUNTS 13-27	28-33 T/A NO. / P.O. NO.	34-39 STATION CODE / PROP. NO. / PROJECT NO.	40-42 EXPEND CODE / FY	43 FUND	45-46 PAY PERIOD CODE	47-52 OBLIG. REF. NO. / ADVANCE ACCT. NO. / EMP. NO.	53 CA YR	54-57 GENERAL LEDGER ACCT. NO.	58-67 ALLOT. OR COST ACCT. NO.	62-67 WORK REF. NO. / X REF. NO.	68-70 OBJECT CLASS	71-80 AMOUNT DEBIT	71-80 AMOUNT CREDIT
[illegible]				40	1		1016		6610	25 1390	3902	732	793 44	
[illegible]			25 [illegible]				1016		1429	/			19 836 00	
[illegible]														793 44
[illegible]												TOTALS		19,836 00

FORM 6-80 281 USE PREVIOUS EDITIONS. (38-49)

143-10

[redacted]

February 10, 1966

[redacted]

Gentlemen:

[redacted] notified us that you have requested a detailed statement of his grant from [redacted]. We have prepared the detailed statement which you require.

Please let us know if there is anything more you need.

Sincerely,

[redacted]

Research Accountant

[redacted]

accounting to BF rx
23 Jan. 1966

143-11

February 23d, 1966

[redacted]

Attention: [redacted]

Dear [redacted]

[redacted] told me he discussed the [redacted] account with you this morning and it was agreed that:

1. You would again contact the contractor re the $18,20 unexpended balance reflected in the accounting for the period ending August 31, 1962, and not carried forward to the next grant year;

2. the current grant period would be extended in time only to 30 June 1966, giving the grantee the opportunity to expend in the field the $123.69 residue, as requested in the January 10th, 1966 letter from [redacted] (the original of this letter is enclosed herewith).

Also enclosed is the original of the February 10th, 1966 letter from the [redacted] which forwarded this grantee's accounting for expenditures during the period January 1st, 1963 through October 22nd, 1965.

Thanks very much for your help in the first item.

Enclosure: (2)

#143

143-12

19 January 1966

MEMORANDUM FOR : Chief, TSD/BB

SUBJECT : Delinquent Accountings
MKULTRA, Subproject 143

1. Review of MKULTRA, Subproject 143 financial records indicate that accountings for grants are now past due. It is requested that your office initiate action to obtain the delinquent accountings.

2. For your information funds have been advanced to the grantee as follows.

Authorization 1 $19,836.00
Grant period 1 Jan. - 31 Dec. 1963

Authorization 2 $20,000.00
Grant period 1 Jan. - 31 Dec. 1964

[redacted]
Chief, TSD/SS

Distribution:
Original & 1 - Addressee

[redacted]

[redacted]

143-13

Receipt is hereby acknowledged of [redacted]
dated 10 February 1964
drawn on the [redacted]
payable to [redacted]
in the amount of
$20,000.00.

Receipt is hereby acknowledged of [redacted]
dated 7 February 1964,
drawn on the [redacted]
payable to [redacted]
in the amount of $800.00

[redacted]
Secretary

Date Feb. 14, 1964

[redacted] MKULTRA 143-14

PROJECT CRYPTO : MKULTRA #143

GRANTEE : [redacted] B

PRINCIPAL INVESTIGATOR : [redacted] C

DATE RENEWED : 1 January 1964

During the past year, the investigators have studied a total of 57 compounds for their capacity to stimulate bacteria. These 57 compounds consisted of 34 phenols, 13 amines, 5 aldehydes and ketones, 3 quaternary ammonium compounds and 2 unclassified chemicals. This study has shown that the phenolic chemicals are the best compounds for stimulating bacteria. Of this group they have noted that Q-phenyl-phenol, p-tert-butylphenol and nonyl phenol are the best compounds studied so far.

They have been able to produce as much as a four-fold increase in microbial growth using these compounds. This degree of stimulation is not as great as they had hoped it would be. For this reason, during the past three weeks they changed their line of approach to the problem.

They have produced a product consisting of phenol, casein hydrolyzate, yeast extract and peptone which has a much greater stimulating capacity. It appears that concentrations as small as 50 ppm will give them a consistant 10 fold increase in growth. The deterioration of the cutting fluid is greater in three days than they have ever observed in their studies with these products at any time. In addition, as yet unpublished information from this laboratory indicates that this produce should neutralize practically any known biocide that might be added to cutting fluids.

It is believed that they have arrived at a point where dynamic progress can be anticipated. In view of the high proudcitivity and promise of results that will be of direct benefit to the operational aspects of DDP. It will be the purpose of the renewed project to combine the two principles of bacterial stimulation (the use of subeffective concentrations of inhibitors and the use of non-specific protective compounds) to determine the ultimate stimulation to be expected in cutting fluids. More emphasis will be given to actual degradative processes rather than stimulation of growth per se. In addition [redacted] has agreed to investigate microbial fouling of diesel cooling systems in conjunction with his other work. C

[redacted] MKULTRA

(When Filled In)

REQUEST FOR ADVANCE OF FUNDS

NAME OF ADVANCEE	DATE	AMOUNT	VOUCHER NO. (Finance use only)
[redacted]	30 January 1964	$800.00 $20,000.00	
PAYABLE TO	ROOM / BUILDING	TELEPHONE EXT.	
See attached payment inst. E-10	[redacted]	[redacted]	

PURPOSE:

Advance to [redacted] for funding MKULTRA Subproject 143, invoice # 2 which activity was approved by C/TSD on 25 January 1964. Accounting to be in accordance with MKULTRA Fiscal Annex attachment B.

Please forward checks (2) to C/TSD/BB thru TSD/BF no later than 10 February 1964.

STATUS OF OUTSTANDING ADVANCES

DATE	AMOUNT	UNACCOUNTED BALANCE

I agree that I will fully account for this advance by submission of vouchers and refund of any unexpended balance to the reporting point stated and by the due date checked below. In the event of my failure to so account and refund any unexpended balance, I authorize deduction from my salary to effect settlement.

REPORTING POINT	DUE DATE	✓
FINANCE DIVISION - HEADQUARTERS		
	ON ARRIVAL AT DESTINATION ON OR ABOUT	
	MONTHLY - ON THE LAST WORKDAY OF EACH MONTH	

REQUESTING OFFICER

DATE: 1/30/64 SIGNATURE: [redacted]

I CERTIFY FUNDS ARE AVAILABLE

OBLIGATION REFERENCE NO.: CHARGE ALLOTMENT NO.: 4125-1390-3202

DATE: 4 FEB 1964 SIGNATURE:

APPROVED

DATE: A SIGNATURE OF APPROVING OFFICER: SIDNEY GOTTLIEB, DC/TSD

CERTIFIED FOR PAYMENT OR CREDIT

DATE: AUTHORIZED CERTIFYING OFFICER:

RECEIPT FOR FUNDS ADVANCED

I acknowledge receipt of funds in the amount stated hereunder to be used for the purpose stated and accounted for as shown above.

DATE: AMOUNT:

SIGNATURE OF ADVANCEE:

SPACE BELOW FOR EXCLUSIVE USE OF FINANCE DIVISION

PREPARED BY: REVIEWED BY: VOUCHER NO. 7-12

DESCRIPTION-ADVANCE ACCOUNTS 13-27	DESCRIPTION-ALL OTHER ACCOUNTS 13-33	28-33 T/A NO. P.O. NO. SF-30 DIV.	34-39 STATION CODE PROP. NO. PROJECT NO.	40-42 EXPEND CODE FY	43 FUNDS	45-46 PAY PERIOD CODE	47-52 OBLIG. REF. NO. ADVANCE ACCT. NO. EMP. NO.	53 CAL YR	54-57 GENERAL LEDGER ACCT. NO.	58-67 ALLOT. OR COST ACCT. NO.	62-67 CH. NO. X REF. NO.	68-70 DUE DATE OBJECT CLASS	71-80 AMOUNT DEBIT	71-80 AMOUNT CREDIT
											TOTALS			

FORM 5-60 281 USE PREVIOUS EDITIONS.

* U.S. GOVERNMENT PRINTING OFFICE : 1962 O—649743

(38-49)

14-52-16

Date 8 January 1964

Branch BB Category COVERT ACTION SUPPORT Sabotage (IVa)

Project Title _____ Item Classification Unclassified (Data)

Project Crypto MKULTRA 143 Crypto Classification Unclassified

Branch Project No. N.A. Project Engineer [redacted] A

Contractor [redacted] C

Contract No. N.A. Task No. N.A.

Type of Contract MKULTRA Date Initiated 1 January 1964

Cost FY 63 - $20,629.44
FY 64 - $20,800.00 Completion Date 1 year

Purpose: The objective of the first year was achieved. An approach to the stimulation of soluble-oil degradation was proved feasible. In addition a system for negating microbial inhibitors was developed. These two accomplishments appear to be the first significant breakthrough for microbial degradation of materials. It is planned to study further the limitations of these approaches and their application to a variety of practical situations during the coming year.

Status: Performance has been excellent. Has progressed beyond expectations. Current in fiscal matters.

Requirement: Generated by undocumented requests from DDP units and divisions as well as necessity for TSD to maintain a quick delivery capability to meet anticipated future operational needs.

[redacted]

4. The cost of this program for the second year is $20,000.00. To this sum must be added $800.00 representing a four per cent service charge to the [redacted] } B [redacted] The [redacted] will function as cut-out and cover for this grant. The cover title of the project is [redacted] -C

5. Total cost of the project will not exceed $20,800.00 Charges should be made against Allotment No. 4125-1390-3902.

6. The [redacted] will request the [redacted] B to submit a summary accounting of monies received and to return any unexpended grant funds at the end of the project year.

7. Title to any permanent equipment purchased by funds granted the [redacted] shall be retained by them in lieu of higher overhead rates. B

8. It was mutually agreed that documentation and accounting for travel expenses which are reimbursable by the [redacted] hall conform with the accepted practices of -B that institution. -A

[redacted]
TSD/Biological Branch

Attachments:
Proposal & budget

Note: Due date for accounting of previous year's funds is 31 March 1964.

Distribution:
Original only

143-17

[redacted]/MKULTRA

DRAFT

8 January 1964

MEMORANDUM FOR : THE RECORD

SUBJECT : MKULTRA, Subproject 143

1. The purpose of subproject 143 is to continue support of the work of [redacted] at the [redacted] on an approach to microbial deterioration of materials.

2. During the first year a system was developed which allows a 10 fold stimulation of the microbial deterioration of soluble oil coolants (as a type system) using extremely small quantities of active material. In addition a system was found which apparently serves to inactivate most microbial inhibitors in such a product. This was much more of an accomplishment than was expected. For the coming year it is planned to explore the limitations of these approaches as they apply to cutting oils and other petroleum based products. It is hoped that these techniques will be applicable across-the-board to all sabotage or harassment systems involving microorganisms, promoting a more deliverable package to the operator.

3. [redacted] principal investigator, has been cleared through SECRET.

[redacted]/MKULTRA

[redacted] 143-18

Date: 8 January 1964

MEMORANDUM FOR : C/TSD/FASS

SUBJECT : MKULTRA, Subproject No. 143

Under the authority granted in the memorandum dated 13 April 1953 from the DCI to the DD/A, and the extension of this authority in subsequent memoranda, Subproject 143 has been approved, and $20,800.00 of the over-all Project MKULTRA funds have been obligated to cover the subproject's expenses and should be charged to cost center 4125-1390-3902

A

[redacted]
PROJECT MANAGER

Concur:

[redacted]
13 JAN 1964

[redacted]
400
4125-1390-3901

[redacted]
AUTHORIZING OFFICER

APPROVED FOR OBLIGATION OF FUNDS

[redacted]
DD/A

Date: [redacted]

Distribution:
Orig & 1 - Addressee
2 - TSD, BB

GROUP 1
Excluded from automatic downgrading and declassification

143-19

[redacted]

B

January 3, 1964

[redacted]

A

Dear [redacted]

As you know, the grant from the [redacted]s B one year old and it is time to consider renewal of the grant.

During the past year we have studied a total of 57 compounds for their capacity to stimulate bacteria. These 57 compounds consisted of 34 phenols, 13 amines, 5 aldehydes and ketones, 3 quaternary ammonium compounds and 2 unclassified chemicals. This study has shown that the phenolic chemicals are the best compounds for stimulating bacteria. Of this group we have noted that O-phenyl-phenol, p-test-butylphenol, di-sec-butylphenol, dodecyl pehnol, 2,4-dimethyl-6-test-butylphenol and nonyl phenol are the best compounds studied so far.

We have been able to produce as much as a four-fold increase in microbial growth using these compounds. This degree of stimulation is not as great as we had hoped it would be. For this reason, during the past three weeks we changed our line of approach to the problem.

We have produced a product consisting of phenol, casein, hydrolyzate, yeast extract and peptone which has a much greater stimulating capacity. It appears that concentrations as small as 50 ppm will give us a consistant 10 fold increase in growth. The deterioration of the cutting fluid is greater in three days than we have ever observed in our studies with these products at any time. In addition, as yet unpublished information from this laboratory indicates that this product should neutralize practically any known biocide or stat that might be added to cutting fluids.

We feel that we now have arrived at a point where we can make some dynamic progress. During the coming year we hope to:

1. Study additional chemicals and media products for their stimulatory capacity.
2. Determine the best ratios of the media components and generally improve the formulation.
3. Determine the effect of the formulation on other petroleum products.
4. Determine if the formulation has the capacity to neutralize anti-microbial agents.
5. Begin work on how the formulation stimulates microorganisms.
6. Determine the effect of the formulation plus urine on microbial growth.

I believe these points cover the main objectives for the coming year.
Kindest personal regards.

Sincerely yours,

[redacted]

C

[redacted]

PROPOSED GRANT COST

A.	Principal Investigator ($\frac{1}{6}$ time)	3990.00
	Full time research assistants	9417.00
B.	Expendable equipment and supplies	2560.00
C.	Typing reports, papers, etc.	700.00
D.	Overhead (20%)	3333.00
	TOTAL	20,000.00

143-20

27 January 1964

MEMORANDUM FOR THE RECORD

SUBJECT : MKULTRA Subproject 143

1. At the request of both Chief and Deputy Chief, TSD [redacted] was contacted for the purpose of funding this MKULTRA Subproject through one of the established funding mechanisms controlled by [redacted] Due primarily to the funding date, it was not possible to accomplish this at this time.

2. Arrangements will be made with [redacted] to project our funding requirements for MKULTRA Subprojects on a semi-annual basis. With this lead time, [redacted] feels that they may be capable of funding our total requirements.

3. [redacted] will meet with [redacted] to discuss other Subprojects funding mechanisms at 1115 this date.

[redacted]
Chief, TSD/Support Staff

GROUP 1
Excluded from automatic downgrading and declassification

[redacted] C REVISED 143-71

Date 5 September 1963

COVERT ACTION SUPPORT

Branch BB Category Sabotage (IVa)
(formerly IIIb)

Project Title Item Classification Unclassified (Data)

Project Crypto MKULTRA 143 Crypto Classification Unclassified

Branch Project No N.A. Project Engineer [redacted] A

Contractor [redacted] C

Contract No N.A. Task No N.A.

Type of Contract MKULTRA Date Initiated 1 January

Cost $20,629.44 Completion Date 1 year

Purpose: To investigate the potential of a new approach to the stimulation of microbial deterioration of petroleum products and its possible application to other degredative processes.

Status: Contractor has developed a model system for desired system. Will now attempt to apply it on small scale. Current in fiscal matters.

Requirement: Generated by undocumented requests from DDP units and divisions as well as necessity for TSD to maintain a quick delivery capability to meet anticipated future operational needs.

[redacted] [redacted]

143-22

RECEIPT

Receipt is hereby acknowledged of [redacted] E
[redacted] dated January 7, 1963, drawn on the [redacted] B
payable to [redacted]
in the amount of $19,836.00

Receipt is hereby acknowledged of [redacted] E
[redacted] dated January 7, 1963, drawn on the [redacted] B
payable to [redacted]
in the amount of $793.44

[redacted] C

Date Jan 10 1963

[redacted] 143-23

DRAFT

11 December 1962

MEMO FOR : THE RECORD

SUBJECT : MKULTRA, Subproject 143

1. The purpose of subproject 143 is to support the work of [redacted] C & B on an approach to the stimulation of petroleum product deterioration.

2. Although the model system for these studies will be petroleum products, this appraoch to stimulation of deterioration may have wide application to all areas of biological sabotage currently of interest to TSD.

3. [redacted] principal investigator, has been cleared C through SECRET.

4. The cost of this program for the first year is $19,836.00. To this sum must be added $793.44 representing a four per cent service charge to the [redacted] will B function as cut out and cover for this grant. The cover title of the project is [redacted] } C

5. Total cost of the project for a period of one year will not exceed $20,629.44. Charges should be made against Allotment No.

[redacted]

3125-1390-3902.

6. [redacted] will request the [redacted] to submit a summary accounting of monies received and to return any unexpended grant funds at the end of the project year.

7. Title to any permanent equipment purchased by funds granted the [redacted] shall be retained by them in lieu of higher overhead rates.

8. It was mutually agreed that documentation and accounting for travel expenses which are reimbursable by the [redacted] shall conform with the accepted practices of that institution.

9. The requirement for a six months' informal accounting on the part of the principal investigator is waived.

[redacted]
TSD/Biological Branch

[redacted]
TSD/Biological Branch

Attachment:
Proposal & Budget

Distribution:
Original only

[redacted]

148-23

GRANT REQUEST

from

[redacted]

C & B

Biographical Sketch

Entire page C

Publications and Papers:

5 pages C

Symposia and Books

1 page

1413-23

B

FACILITIES

[redacted] and can place the following equipment at the disposal of this project:

Balances (Sartorius, etc.)

Warburg

Ultra-centrifuge

Ratheon Sonic Oscillator

Mickle Disintegrator

Centrifuges(refrigerated, Sorval, International, Ultra-centrifuge, Spinco, etc.)

Sharples Super Centrifuge

Beckman DU 60 Spectrophotometers

Beckman B

Coleman Photofluorometer

Refrigerated Warburg

Incubators

Research pH Meters

Shaking Incubators

Kjeldahl Apparatus

Colorimeters

Pilot Plant Fermentors

Chromatography Equipment

Fraction collectors

Isotope laboratory with complete supporting equipment

OBJECTIVE OF THIS INVESTIGATION

To determine if the growth of microorganisms in petroleum products can be markedly increased by the addition of substatic concentrations of antimicrobial agents.

LITERATURE REVIEW

The literature has been carefully searched to determine what is known concerning the use of antimicrobial agents to stimulate microbial growth. To date, only two papers have been found that relate to this topic.

Imshenetskii and Perova (Institute of Microbiology, Academy of Science U.S.S.R., Mikrobiologiya, 23, 53-56, 1954) found that cultures of Saccharomyces cerevisiae trained to grow in the presence of small concentrations of $HgCl_2$ or phenols have much higher fermentation rates than cultures that have not been trained to grow in the presence of these compounds. As little as 50 ppm of $HgCl_2$ accelerates the production of beer. These workers believe that this effect is due to traces of inhibitor in the environment and not because of some change in the physiology of the cells.

[redacted] reported that 50 ppm of 2,4,6-trichloro phenol could markedly stimulate the growth of bacteria in emulsion oils. In some instances this compound produced almost a three fold increase in the number of cells in the environment.

[redacted] found that 25 to 150 ppm of 2,4-dinitrophenol increases the oxidation of C_4, C_6, C_8, C_{10}, C_{12}, C_{13}, C_{14}, and C_{16} fatty acids. The increase was as much as 100% and was not associated with an increase in

the number of cells in the environment.

These results imply that two types of stimulation may be produced by substatic concentrations of inhibitors. First, a metabolic stimulation may be produced where the cells do not increase in number but this metabotic rate is increased. Second, the number of cells in the environment may be increased by the compound. It remains to be seen whether there is a stimulation in the individual metabolic activity of each cell in this case. There is an increase in activity, however, this may only be due to the greater number of cells.

UNPUBLISHED DATA

B

[redacted] have found that 1000 ppm of 2-nitro-1-butyl acetate produce as 3 fold increase in the production of penicillin by Penicillium chrysogenum.

C

PLAN OF ATTACK

It is well known that the rate of oxidation of hydrocarbons may be limited by the rate of dufision of the compound across the permeability barrier. It is also well known that surface active agents increase the permeability of the cytoplasmic membrane. Several years ago [redacted] developed the idea that the use of surface active compounds may increase the accessibility of hydrocarbon substrates to microbial attack and thus speed up the oxidation of these compounds.

Due to several factors including a lack of funds, this idea has never been tested to any extent. To date, we have studied 3 phenolic compounds. Two of these compounds produced stimulation and one did not.

We propose to study the effect of substatic concentrations of ph

phenols, fatty amines/alcohols and quatemary ammonium compounds on the oxidation of two different synthetic coolants and one petroleum type product. We may also use other antimicrobial agents other than those listed above.

EXPERIMENTAL PROCEDURE

CULTURES

Pure cultures of microorganisms belonging to the Genus Pseudomonas will be used in this investigation. The cultures will be isolated from samples of used cutting fluids obtained in the Houston area.

MEDIA

Whenever possible, the organisms will be maintained in cutting fluids. In some instances conventional media such as nutrient broth, meat infusion broth, or an inorganic salts medium may be used in this investigation.

INCUBATION TEMPERATURE

All experiments will be done at 30°C. The length of the incubation period will vary according to the organism and the particular experiment.

CHEMICALS

The best grade of chemical obtainable will be used throughout this investigation.

GLASSWARE

All flasks, tubes and other glass equipment will be acid cleaned prior to each experiment.

HYDROGEN ION CONCENTRATION

The pH of the environment will vary according to the particular experiment. In general, we will let the Ph of the cutting fluid

determine the pH of the experiment.

INOCULUM

The cells used in the experiments will be grown on an appropriate medium. They will be harvested after a suitable incubation period and washed three times in isotonic saline. The cells will be suspended in isotonic saline and uniform quantities of this suspension will be used to inoculate the tubes or flasks.

STERILITY

We will employ suitable procedures for sterilizing the media containing the petroleum products.

CONTROLS

Appropriate controls will be included in each experiment. Each experiment will be done in duplicate and repeated at least once.

CUTTING FLUIDS

Three different cutting fluids will be used in this investigation consisting of two chemical coolants and one petroleum product.

EXPERIMENTAL TECHNIQUE

Each cutting fluid will be prepared in a 1-25 dilution in tap water. One hundred ml of this coolant will be placed in each 250 ml Erlenmeyer flask. The number of flasks used in each experiment will vary from time to time. The inhibitor will be added to the series of flasks in varying concentrations. These flasks will be sterilized by autoclaving and then inoculated with the test organisms. The flasks will be incubated for a period of time (probably two weeks) in a shaking incubator.

At periodic intervals of about 2 days time the number of cells per ml will be determined for each flask. At the same time the pH of the flask will be determined and other biochemical tests (such as dehydrogenase activity) may be conducted to determine the degree of metabolic activity in the environment. At the end of the experiment

the remaining cutting fluid will be poured into a volumetric cylinder to determine the quantity of free oil in the product (this test will be used on the petroleum coolant only). The quantity of free oil present is an excellent index of the degree of attack on the emulsifier.

In addition, Warburg studies will be conducted at the optimim stimulatory concentration determined in the growth expedent to establish the effect of the antimicrobial agent on oxygen uptake.

We will study as many compounds as possible during the year period of this grant. In no case will this be less than twenty-five compounds.

COST OF PROJECT

A.	Principal Investigator $200.00/month (9 months)	$ 1,800.00
	Full time salary to work on project in Summer	3,210.00
	2 Full Time Research Assistants ($375/month each)	9,000.00
		$ 14,010.00
B.	Permanent Equipment	
	1 Shaking incubator	$ 1,080.00
C.	Expendable Equipment & Supplies	$ 900.00
D.	Publication Cost	
	Typing report, papers, and purchasing reprints of publications that may arise from this work	$ 400.00
		$ 16,390.00
E.	Overhead (20%)	$ 3,446.00
		$ 19,836.00